

Lecture 4

Example-8

Example- An unloaded zener voltage regulator has an input source voltage of 20 V, a series resistance of $330\ \Omega$, and a zener voltage of 12 V. How much current is flowing through the zener diode ?

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Solution

$$V_i = 20 \text{ V}, R = 330 \Omega, V_Z = 12$$

Since load is disconnected, $I_L = 0$. Therefore,

$$I_Z = I_R = \frac{V_i - V_Z}{R} = \frac{20 - 12}{330} = 24.2 \text{ mA}$$

Example-9

Example- An unloaded zener voltage regulator has an input source voltage of 20 V, a series resistance of $330\ \Omega$, and a zener voltage of 12 V. If the series resistor has a tolerance of $\pm 10\%$, what is the maximum zener current ??

Example-9

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Solution

$$V_i = 20 \text{ V}, R = 330 \Omega, V_Z = 12$$

Minimum value of series resistor, $R_{\min} = 330 - 0.1 \times 330 = 297 \Omega$. Thus, maximum value of zener current is

$$I_Z = I_R = \frac{V_i - V_Z}{R_{\min}} = \frac{20 - 12}{297} = 26.9 \text{ mA}$$

Example-10

- Design the network of Fig. 12 to maintain V_L at 12 V for load variation (I_L) from 0 mA to 200 mA. That is, determine R_S and V_Z .
- Determine P_Z max for the Zener diode of part (a).

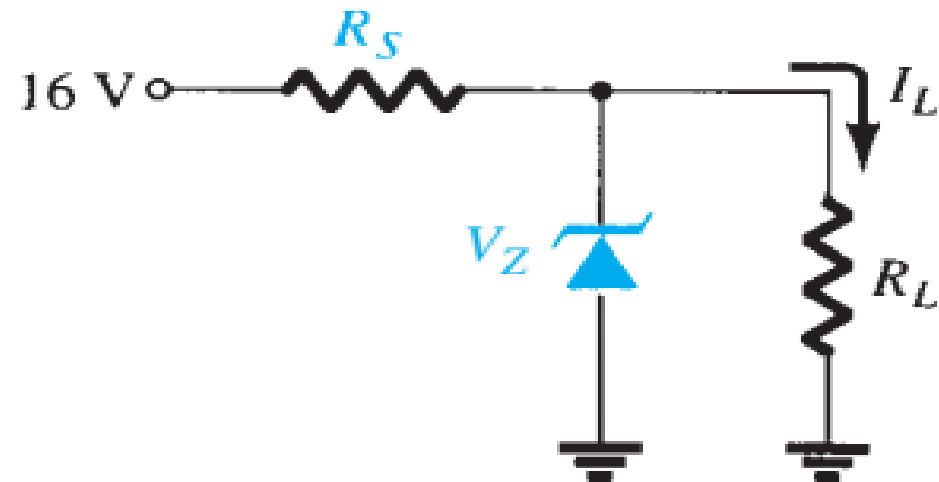


Fig. 12

Example-10

- Design the network of Fig. 12 to maintain V_L at 12 V for load variation (I_L) from 0 mA to 200 mA. That is, determine R_S and V_Z .
- Determine P_Z max for the Zener diode of part (a).

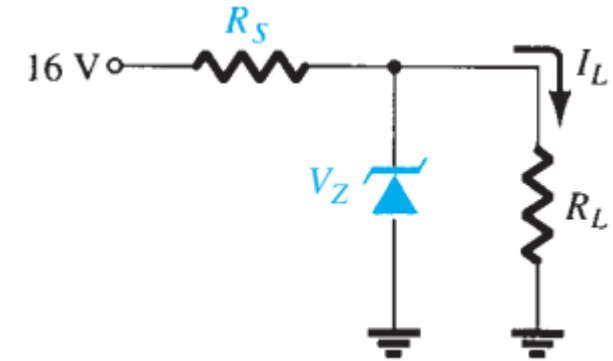


Fig. 12

Solution (a)

Given, $V_L = 12\text{V}$, $I_{L\min} = 0$, $I_{L\max} = 200\text{ mA}$.

The **Zener diode must be on** to maintain V_L at 12V and V_Z must also be equal to V_L . Therefore,

$$V_L = V_Z = 12\text{V}$$

The **minimum value of load resistance to make the Zener diode on** is given by

$$R_{L\min} = \frac{R_S V_Z}{V_i - V_Z} = \frac{R_S (12)}{16 - 12} = 3R_S$$

Cont...

The load current is maximum when load resistance is minimum and given by

$$I_{L\max} = \frac{V_Z}{R_{L\min}}, \quad \Rightarrow \quad 200 \text{ mA} = \frac{12}{3R_S},$$

$$\mathbf{R_S = 20\Omega}$$

Once the Zener diode is in the “on” state, the voltage across R_S remains fixed at

$$V_{RS} = V_i - V_Z = 16 - 12 = \mathbf{4V}$$

The current through the R_S is also fixed and given as

$$I_R = \frac{V_{RS}}{R_S} = \frac{4}{20} = 0.2A$$

Cont...

The current through the Zener diode is maximum (I_{ZM}) when the load current is minimum and given as

$$I_{ZM} = I_R - I_{Lmin} = 0.2 - 0 = 0.2A$$

Solution (b)

The maximum power rating of Zener diode is given as

$$P_{ZM} = I_{ZM} V_Z = 0.2(12) = 2.4 \text{ W}$$

Example-11

EXAMPLE The network of Fig. 13 is designed to limit the voltage to 20 V during the positive portion of the applied voltage and to 0 V for a negative excursion of the applied voltage. Check its operation and plot the waveform of the voltage across the system for the applied signal. Assume the system has a very high input resistance so it will not affect the behavior of the network.

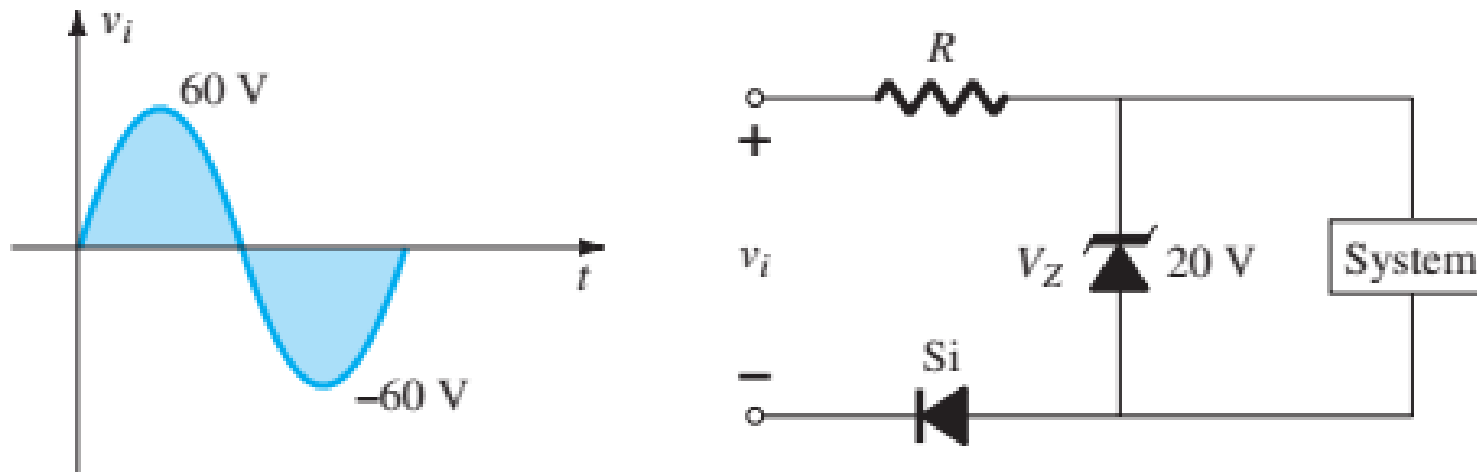


Fig. 13

Example-11

EXAMPLE The network of Fig. 13 is designed to limit the voltage to 20 V during the positive portion of the applied voltage and to 0 V for a negative excursion of the applied voltage. Check its operation and plot the waveform of the voltage across the system for the applied signal. Assume the system has a very high input resistance so it will not affect the behavior of the network.

Solution

Once the voltage across the Zener diode reaches 20 V the Zener diode will turn on as shown in Fig. 14 and the voltage across the system will lock in at 20 V. Further increases in the applied voltage will simply appear across the series resistor with the voltage across the system and the forward-biased diode remaining fixed at 20 V and 0.7 V, respectively.

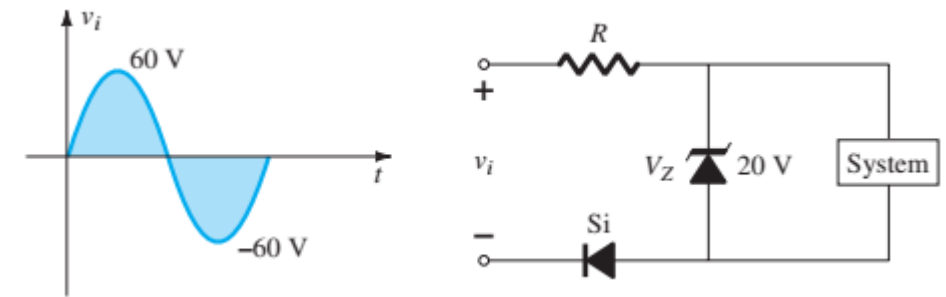


Fig. 13

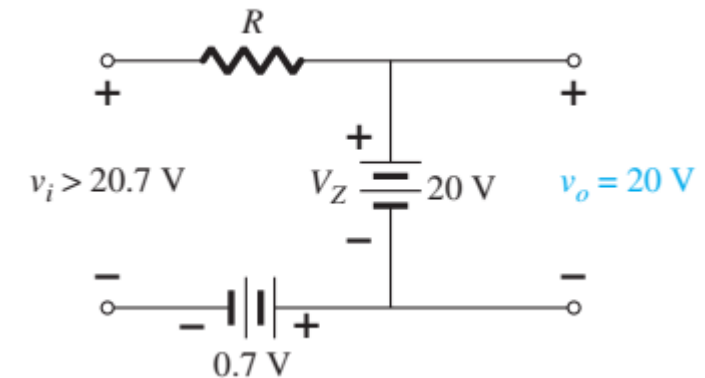


Fig. 14

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For the negative region of the applied signal the silicon diode is reverse biased and presents an open circuit to the series combination of elements. The result is that the full negatively applied signal will appear across the open-circuited diode and the negative voltage across the system locked in at 0 V, as shown in Fig. 15.

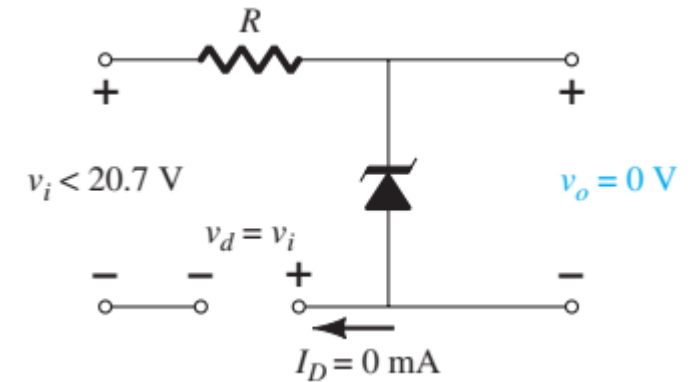


Fig. 15

The voltage across the system will therefore appear as shown in Fig. 16.

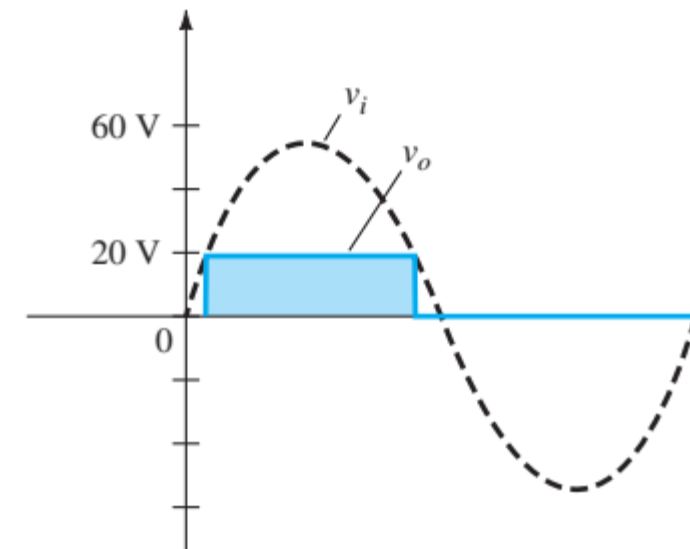


Fig. 16

References

1. Robert L. Boylestad, Louis Nashelsky, “Electronic Devices and Circuit Theory”, 11th ed, Prentice Hall of India, 2014.
2. A. S. Sedra,, and K. C. Smith, ‘Microelectronic Circuits’. Oxford Oxford University Press, 7th Edition, 2012.